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First measurement of time-dependent *CP* violation in the flavor-changing neutral-current decay $B^0 \to K^0_S \mu^+ \mu^-$

LHCb collaboration[†]

Abstract

A flavor-tagged time-dependent analysis of $B^0 \to K^0_S \mu^+ \mu^-$ decays is performed across the full dimuon mass range excluding the J/ψ and $\psi(2S)$ resonance regions. The analysis uses proton-proton collision data collected by the LHCb experiment in 2011–2018 at center-of-mass energies of 7, 8 and 13 TeV, corresponding to an integrated luminosity of $9\,\mathrm{fb}^{-1}$. The *CP* violation parameters are determined to be

$$C = -0.13 \pm 0.32 \pm 0.04\,,$$
$$S = +0.82 \pm 0.29 \pm 0.05\,,$$

where the first uncertainties are statistical and the second are systematic. The results are consistent with the Standard Model prediction. This is the first experimental study of time-dependent *CP* violation in $b \to s\ell^+\ell^-$ processes.

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Flavor-changing neutral-current decays [1], such as the rare $b \to s\ell^+\ell^-$ transition, serve as powerful and sensitive probes for physics beyond the Standard Model (SM). Their inherent suppression within the SM makes any observed deviation in their decay properties a clear signature of new physics. Since beginning of operations, the LHCb experiment has performed extensive studies of a series of $b \to s\ell^+\ell^-$ processes, which have revealed a compelling, albeit evolving, picture of anomalies. Most notably, the measurements of branching fractions and angular observables in several semileptonic B-meson decays [2–14] show a coherent and distinct pattern of tensions with the SM, which could be explained by new-physics contributions to the effective four-fermion $b \to s\ell^+\ell^-$ couplings [15–17].

These persistent anomalies motivate the use of complementary experimental probes to better understand the underlying dynamics. In particular, time-dependent analyses of neutral B-meson decays provide theoretically clean observables to explore new-physics phenomena in $b \to s$ transitions [18–27]. The CP asymmetry arising from the interference between the B^0–$\overline{B}^0$ mixing and decay amplitudes is sensitive to the imaginary parts of the Wilson coefficients governing these processes, yielding constraints on new CP-violating phases that are complementary to, and in many cases more direct and robust than, those obtained from the time-integrated or time-independent decay rates, angular distributions and CP asymmetries in decays such as $B^0 \to K^*(892)^0\mu^+\mu^-$ [28–30]. The power of the time-dependent approach is well established in hadronic $b \to sq\bar{q}$ $(q = d, s)$ channels, such as $B^0 \to \eta' K^0_S$ and $B^0 \to \phi(1020)K^0_S$ [31, 32].[1] The decay $B^0 \to K^0_S\mu^+\mu^-$ thus enables the first time-dependent CP violation measurement in a $b \to s\ell^+\ell^-$ process, opening a crucial new window to CP-violating new physics in these rare modes.

In general, the $K^0_S\mu^+\mu^-$ final state is an admixture of CP-even and odd eigenstates, thus a full time-dependent angular analysis is required to perform the most complete test of CP violation [19, 33]. Given the limited statistics of the data sample considered in this Letter, this study focuses on measuring the overall time-dependent CP asymmetry after integrating over the lepton helicity angle in the dimuon rest frame. The time-dependent CP asymmetry between the decay rates of initial $\overline{B}^0$ and B^0 mesons to the $K^0_S\mu^+\mu^-$ final state is defined as

$$\mathcal{A}^{CP}(t) \equiv \frac{\Gamma(\overline{B}^0(t) \to K^0_S\mu^+\mu^-) - \Gamma(B^0(t) \to K^0_S\mu^+\mu^-)}{\Gamma(\overline{B}^0(t) \to K^0_S\mu^+\mu^-) + \Gamma(B^0(t) \to K^0_S\mu^+\mu^-)} \simeq S\sin(\Delta m_d t) - C\cos(\Delta m_d t), \tag{1}$$

where t is the time interval between the production and decay of the $\overline{B}^0$ or B^0 meson in its rest frame, Δm_d is the B^0–$\overline{B}^0$ oscillation frequency, and Γ is the decay rate for corresponding processes. The decay-width difference of the two B^0 mass eigenstates, $\Delta\Gamma_d$ is assumed to be zero in the equation. The parameters S and C quantify the mixing-induced and direct CP violation, respectively. In the SM, the $B^0 \to K^0_S\mu^+\mu^-$ decay proceeds dominantly to a CP-odd state and the expectation is $C \approx 0$ and $S = \sqrt{(1 - C^2)}\sin(2\beta + \Delta\phi_d) \approx \sin(2\beta)$. Here β is the CKM angle measured in $B^0 \to J/\psi K^0_S$ decays [34], and $\Delta\phi_d$ is a small phase shift due to the interference of the B^0 mixing and $b \to s$ decay amplitudes. For the dimuon mass-squared range $1.1 < q^2 < 6.0\,\mathrm{GeV}^2/c^4$, the mixing-induced CP asymmetry is explicitly predicted to be $S = 0.72 \pm 0.02$ in the SM [35]. However, various new-physics models with new interactions or additional weak phases can lead to a significant deviation of S and C from their SM predictions [36].

This Letter reports the first measurement of the CP-violating parameters in

[1]The inclusion of charge-conjugate processes is implied throughout this Letter unless otherwise stated.

$B^0 \to K^0_S \mu^+\mu^-$ decays using pp collision data collected with the LHCb detector in 2011–2012 (Run 1) and 2015–2018 (Run 2), corresponding to an integrated luminosity of $9\,\text{fb}^{-1}$. The CP-violating observables C and S are measured from a time-dependent analysis in the full range of dimuon masses, q^2, excluding the J/ψ ($8.0 < q^2 < 11.0\,\text{GeV}^2/c^4$) and $\psi(2S)$ ($15.0 < q^2 < 17.5\,\text{GeV}^2/c^4$) regions. The control channel $B^0 \to J/\psi(\to \mu^+\mu^-)K^0_S$ is used to study the difference between the simulation and data and to validate the fit procedure.

The LHCb detector [37,38] is a single-arm forward spectrometer covering the pseudorapidity range $2 < \eta < 5$, designed for the study of particles containing b or c quarks. It employs a dipole magnet with periodically reversed polarity for momentum determination. The detector used to collect the data analyzed in this paper includes a high-precision tracking system consisting of silicon-strip [39,40] and straw drift-tube [40] detectors. Different types of charged hadrons are distinguished using information from two ring-imaging Cherenkov detectors [41]. Muons are identified by a system composed of alternating layers of iron and multiwire proportional chambers, achieving high efficiency and excellent momentum resolution [42,43].

The simulated samples used to study the decay properties of the signal, control, and background channels are generated using the software described in Refs. [44–50]. Simulated signal decays are generated according to the theoretical calculation for $b \to s$ transitions in Ref. [51], and calibrated to match the B^0 kinematic distributions and the particle identification (PID) performance in data [43,50,52].

The online event selection is performed by a trigger [53], which consists of a hardware stage followed by a two-stage software trigger. The hardware trigger uses information from the calorimeter and muon systems. Subsequently, the software trigger imposes requirements on the muon transverse momentum and track reconstruction quality, or on the output of a multibody topological b-hadron trigger [54]. Triggered data further undergo a centralized, offline processing step across the entire LHCb physics program [55].

In the offline analysis, K^0_S candidates are reconstructed from two oppositely charged tracks identified as pions. They are categorized according to the position of the K^0_S decay vertex. For K^0_S mesons that decay inside the vertex locator (VELO), the resulting tracks are classified as long tracks and are reconstructed using hits from both the VELO and the main tracking stations, thereby providing the best momentum resolution. For particles that decay outside the VELO, the tracks are classified as downstream tracks; these are reconstructed using only hits from the tracking stations. The mass of the $\pi^+\pi^-$ pair is required to be within $20\,\text{MeV}/c^2$ of the known K^0_S mass [56]. These K^0_S candidates are then combined with a pair of oppositely charged muons to form B^0 candidates. All tracks and vertices are required to have a good fit quality. Furthermore, PID criteria are applied to ensure that the charged tracks are consistent with the hypothesis of being pions or muons, as required by the decay chain. The PID observables of the charged tracks are calculated using multivariate techniques to combine information from different subsystems [43,50,52]. A kinematic fit is performed to improve the B^0 mass resolution by constraining the mass of the $\pi^+\pi^-$ pair to the known K^0_S mass [56].

This analysis considers specific backgrounds that can contaminate the $B^0 \to K^0_S \mu^+\mu^-$ signal sample. These include $\Lambda^0_b \to \Lambda(\to p\pi^-)\mu^+\mu^-$ decays where the proton is misidentified as a pion, and $B^0 \to D^-(\to K^0_S\pi^-)\pi^+$ and $B^0 \to K^0_S\pi^+\pi^-$ decays where two pions are misidentified as muons. These backgrounds are highly suppressed through mass vetos and PID requirements. Evaluation using simulation samples confirms that their residual

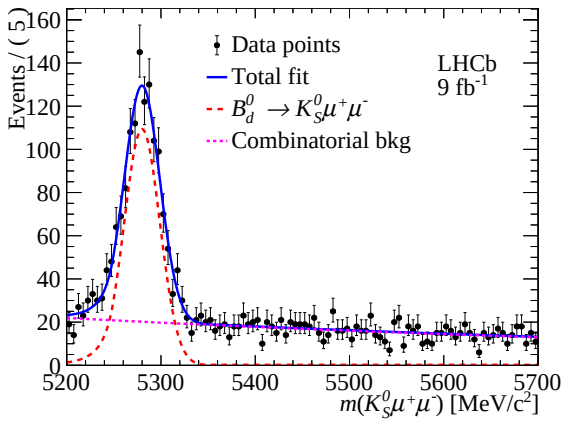

Figure 1: Mass distribution of the selected $B^0 \to K_S^0\mu^+\mu^-$ candidates, with the fit result also shown.

contamination to the final signal sample is negligible after these requirements. The q^2 value is required to be outside the two defined charmonium regions, except when selecting the control channel $B^0 \to J/\psi(\to \mu^+\mu^-)K_S^0$.

To further suppress the combinatorial background arising from the random combination of unrelated tracks, a multivariate analysis based on a gradient boosted decision tree (BDTG) technique is carried out [57]. The BDTG classifier is trained using simulated $B^0 \to K_S^0\mu^+\mu^-$ signal decays and data candidates in the high $K_S^0\mu^+\mu^-$ mass sideband of [5700, 6000] MeV/c^2 to represent the background. The BDTG classifier incorporates a set of kinematic and topological variables, such as the transverse momentum (p_T) and η of the B^0 and K_S^0 candidates, their impact parameters with respect to the primary vertex (PV), and PID variables of the final state particles. These inputs show good agreement between the background-subtracted data and simulation in both the signal and control channels. The requirement on the BDTG output is chosen to maximize the expected signal significance.

Eight categories of $B^0 \to K_S^0\mu^+\mu^-$ candidates are selected using separate BDTG classifiers, corresponding to the combinations of four data-taking periods with roughly equal signal yields (2011–2012, 2015–2016, 2017, 2018) and two K_S^0 categories (with long or downstream track types). An unbinned extended maximum-likelihood fit is performed to the B^0 mass distribution to determine the signal and background yields in each category. The B^0 signal is modeled by a Crystal Ball function [58], with tail parameters fixed from simulation and the width and peak position allowed to vary in the fit. The combinatorial background is modeled with an exponential function.

The mass distribution of the selected B^0 candidates combining all categories is shown in Fig. 1, together with the combined fit results. The total signal yield is 941 ± 55. All selected candidates are then sorted into two groups corresponding to the Run 1 and Run 2 periods, respectively, which are used in the subsequent decay-time fit to determine the CP violation parameters.

The decay-time distribution of B^0 ($\overline{B}^0$) mesons produced in pp collisions is described

by

$$P_{B^0(\overline{B}^0)}(t) \propto e^{-t/\tau_{B^0}} \left[\cosh\left(\frac{\Delta\Gamma_d t}{2}\right) \mp S\sin(\Delta m_d t) \pm C\cos(\Delta m_d t) + D\sinh\left(\frac{\Delta\Gamma_d t}{2}\right)\right], \tag{2}$$

where τ_{B^0} is the lifetime of the B^0 meson. The the upper (lower) sign corresponds to B^0 ($\overline{B}^0$) decays. The parameter D parametrizes CP-violating effects originating from $\Delta\Gamma_d$. Since $\Delta\Gamma_d$ is compatible with zero [56], the $\cosh\frac{\Delta\Gamma_d t}{2}$ term is approximated to 1, and the $\sinh\frac{\Delta\Gamma_d t}{2}$ term in Eq. 2 is neglected in the baseline measurement. The reconstructed decay-time distribution is affected by the experimental effects discussed below.

The flavor of the B^0 or $\overline{B}^0$ meson at production is identified using both the same-side (SS) [59] and opposite-side (OS) [60] flavor tagging algorithms. The SS algorithm exploits charged particles from the signal b-quark fragmentation, such as pions and protons. The OS algorithm utilizes decay products from the associated partner b-hadron, including kaons, leptons, reconstructed charm hadrons, and the charge information from its decay vertex. Each algorithm provides a tagging decision d with an estimated mistag probability, where $d = +1, -1$, or 0 if the meson is classified as B^0, $\overline{B}^0$ or untagged, respectively. The estimated mistag probability is calibrated with a dedicated software package [61], using $B^+ \to J/\psi K^+$ and $B^0 \to J/\psi K^*(892)^0$ decays decays for the OS and SS tagging method, respectively. The calibration samples are weighted to match the distributions in the signal sample of kinematic properties (p_{T}, η, and azimuthal angle of the signal B^0 candidates) and event characteristics (PV and track multiplicities). The package yields the calibrated mistag probability ω for each candidate. The OS and SS tagging decisions and mistag probabilities are then combined to give the final tagging decision and mistag probability for each B meson. The total effective tagging efficiency is $\varepsilon\langle D^2\rangle = (2.7 \pm 1.1)\%$ and $(4.1 \pm 0.6)\%$ for the Run 1 and Run 2 samples, respectively. Here $\varepsilon_{\mathrm{tag}}$ is the fraction of of B candidates that have a nonzero tagging decision, and D^2 is the effective dilution squared depending on the event-by-event mistag probability [59, 60].

Assuming no production or tagging asymmetries for B^0 and $\overline{B}^0$ mesons, the decay-time distribution becomes

$$P(t, d, \omega) \propto [1 + d(1 - 2\omega)]P_{B^0}(t) + [1 - d(1 - 2\omega)]P_{\overline{B}^0}(t)\,. \tag{3}$$

The decay-time resolution is accounted for by convolving Eq. 3 with a resolution function consisting of the sum of three Gaussian functions. Each Gaussian has the same mean and a width depending linearly on the per-candidate decay-time uncertainty. The parameters of the resolution model are obtained by fitting the samples of prompt $K^0_{\mathrm{S}}\mu^+\mu^-$ candidates, which are constructed from a K^0_{S} and two muon candidates compatible with originating at a PV. The resulting resolution model, with an effective resolution of about 60 fs, affects the CP-violating parameters by less than 0.1%, which is negligible compared to the statistical uncertainties.

The reconstruction, trigger, and selection requirements result in a decay-time-dependent signal efficiency. This effect is accounted for by multiplying the decay-time distribution by an acceptance function parameterized by a cubic spline function [62] with six knots at 0.5, 1.0, 1.5, 2.0, 3.0, 8.0 ps and eight coefficients. One coefficient is fixed to unity for normalization, and the remaining coefficients are allowed to vary in the fit to data described below.

A weighted unbinned maximum-likelihood fit [63, 64] is simultaneously performed on the Run 1 and Run 2 data samples to the decay times of the $B^0 \to K^0_{\mathrm{S}}\mu^+\mu^-$ candidates to

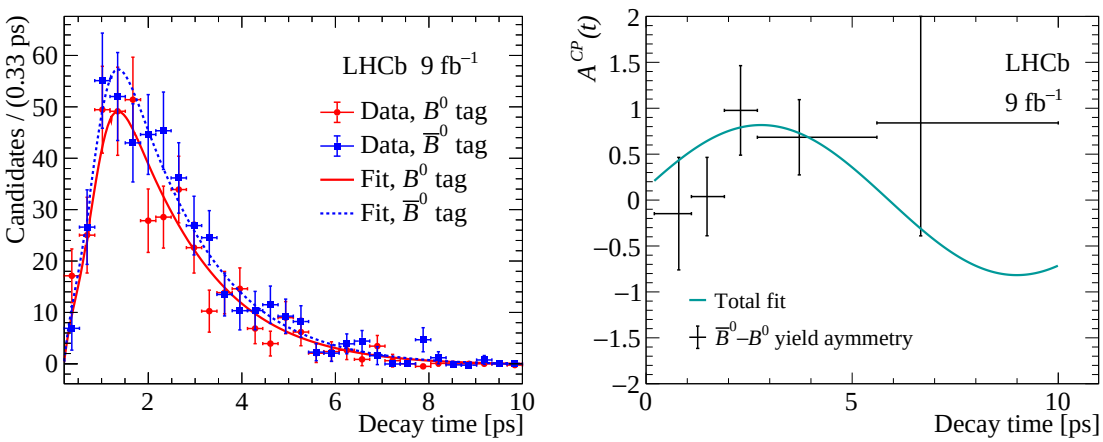

Figure 2: Left: background-subtracted decay-time distributions for $\overline{B}^0$ and B^0 candidates identified through flavor tagging, where the fit result is also shown. The difference between $\overline{B}^0$ and B^0 distributions reveals the *CP* asymmetry. Right: time-dependent *CP* asymmetry. The data points show the binned signal yield asymmetry, with tagging dilution effects corrected for. The result of the maximum-likelihood fit is shown as the solid curve.

determine the *CP* violation parameters S and C. In this fit, only the signal component is explicitly modeled while the background contributions are statistically subtracted via the signal weights calculated with the *sPlot* [65] method using the B^0 mass as the discriminating variable. This method assumes that the decay time and B^0 mass are uncorrelated in both the signal and individual background components. Potential correlations are treated as systematic uncertainties.

In the fit, the B^0 lifetime and the mass difference between the two neutral B mass eigenstates are constrained to their world-average values, $\tau_{B^0} = 1.517 \pm 0.004\,\text{ps}$ and $\Delta m_d = 0.507 \pm 0.002\,\text{ps}^{-1}$ [56]. The decay-time resolution and flavor-tagging calibration parameters are constrained to the fit results obtained in the control channels.

The background-subtracted decay-time distributions of the tagged $\overline{B}^0$ and B^0 signal candidates in the combined data sample are shown in Fig. 2, together with the baseline fit using Eq. 2. The corresponding time-dependent *CP* asymmetry is also presented, which is computed from the tagged yields after correcting for the effects of flavor-tagging dilution. The *CP* violation parameters are determined from the time-dependent fits both for the entire kinematically allowed q^2 range (excluding the J/ψ and $\psi(2S)$ regions) and, to probe potential q^2 dependency, in the q^2 ranges below and above the J/ψ region. The results are summarized in Table 1 and Fig. 3. The 68.3%, 95.5% and 99.7% confidence regions under the physical constraint $C^2 + S^2 \leq 1$ are further obtained for the total q^2 region using the Feldman–Cousins method [66, 67] and shown in the End Matter.

Several sources of systematic uncertainties for the *CP*-violation observables are investigated and listed in Table 2. A large ensemble of pseudoexperiments generated according to the baseline fit results is used to validate the fit procedure. A bias in the C and S parameters is observed, which is due to the limited data sample size. The consistency of the tagging calibration parameters between the control and signal channels is tested using simulated samples, and that between B^0 and $\overline{B}^0$ mesons is investigated using the control channels $B^+ \to J/\psi K^+$ and $B^0 \to J/\psi K^{*0}$. The observed differences represent the leading

Table 1: Summary of the CP violation parameters from the time-dependent fit to $B^0 \to K^0_S \mu^+\mu^-$ decays in the total, low and high q^2 ranges, excluding the charmonium regions. The uncertainties are statistical.

q^2 range $[\mathrm{GeV}^2/c^4]$	C	S
Total	-0.13 ± 0.32	$+0.82 \pm 0.29$
Low: $q^2 < 8.0$	$+0.33 \pm 0.44$	$+1.10 \pm 0.43$
High: $11.0 < q^2 < 15.0$ or $q^2 > 17.5$	-0.62 ± 0.47	$+0.48 \pm 0.47$

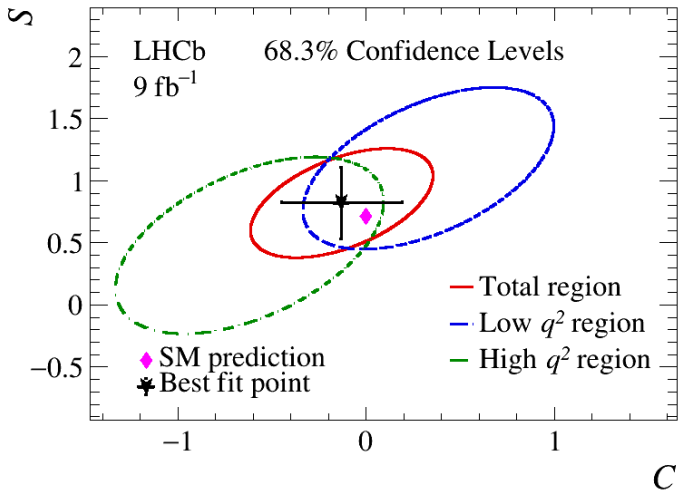

Figure 3: Two-dimensional confidence regions for the parameters C and S from profile likelihood scans, without including systematic uncertainties. The green dash-dotted, blue dashed and red solid contours represent the 68.3% confidence regions for the low-q^2, high-q^2 and total regions, respectively. The black marker with error bars represents the fit results in the total sample, while the magenta marker stands for the SM prediction.

source of systematic uncertainties. Systematic uncertainties associated with the modeling of the signal and background contributions are evaluated by repeating the mass fit with alternative functions, namely the sum of a Student's t-distribution and a Gaussian function for the signal and a second-order polynomial for the combinatorial background. Then, the weighted maximum-likelihood fit is performed, and the differences in the resulting parameters from those of the baseline fit are assigned as systematic uncertainties. The effect of neglecting a small correlation between the reconstructed B^0 mass and decay time in the signal weight determination when applying the *sPlot* method is studied by reevaluating the signal weights in bins of the decay time and performing the decay-time fit again. The effects of a nonzero B^0–$\overline{B}^0$ production asymmetry and a nonzero $\Delta\Gamma_d$ are evaluated by varying their values within their uncertainties [56, 68] and performing the fits again. The resulting largest variations in CP violation parameters are assigned as systematic uncertainties. The systematic uncertainties associated with the decay-time acceptance and resolution models are evaluated using splines with different knot numbers and positions and found to be negligible. The total systematic uncertainties, evaluated as the quadrature sums of the contributions discussed above, are significantly smaller than the statistical ones.

Table 2: Summary of absolute systematic uncertainties for the parameters C and S.

Sources	C	S
Fit bias	0.017	0.015
Flavor tagging	0.031	0.037
Mass fit models	0.010	0.011
Mass-time correlation	0.013	0.018
Nonzero $\Delta\Gamma_d$	0.007	0.005
Production asymmetry	0.006	0.004
Time acceptance and resolution	negligible	negligible
Total	0.041	0.046

The robustness of the analysis results is checked by splitting the data sample by K^0_S reconstruction categories, magnet polarities, and data-taking periods, by varying the B^0 mass and decay-time ranges, and by obtaining alternative determination of the C and S parameters from the time-integrated CP asymmetry. The results are all compatible with each other and with the baseline fit results. As a cross-check, the analysis procedure has also been applied to the control channel $B^0 \to J/\psi K^0_S$. The corresponding results are consistent with those reported in Ref. [34].

In conclusion, this Letter reports a measurement of the time-dependent CP asymmetry in $B^0 \to K^0_S \mu^+\mu^-$ decays using data collected by the LHCb experiment during the Run 1 and Run 2 data-taking periods. The CP violation parameters in the full q^2 range, excluding the dimuon resonant regions due to J/ψ and $\psi(2S)$ states, are determined to be

$$C = -0.13 \pm 0.32\,(\text{stat}) \pm 0.04\,(\text{syst})\,,$$
$$S = +0.82 \pm 0.29\,(\text{stat}) \pm 0.05\,(\text{syst})\,,$$

with a statistical correlation coefficient of 0.50. This is the first measurement of time-dependent CP violation in $b \to s\ell^+\ell^-$ transitions, and the results are consistent with the SM prediction. This study demonstrates a new avenue to probe physics beyond the SM in $b \to s\ell^+\ell^-$ transitions, complementary to measurements of branching fractions, angular observables and lepton-flavor universality observables.

End Matter

1 Confidence regions within the physical boundary

The results shown in Fig. 3 are obtained without applying any physical constraint on the CP violation parameters. In fact, the normalization of the coefficients of the time-dependent decay rate requires $C^2+S^2 \leq 1$. Confidence intervals satisfying this requirement are derived using a constrained likelihood-ratio test. Specifically, the maximum-likelihood estimation and the ordering of parameter points are performed within the restricted domain defined by this inequality. The results are presented in Fig. 4. Systematic uncertainties are excluded from this evaluation as their contribution is negligible compared to the statistical uncertainties.

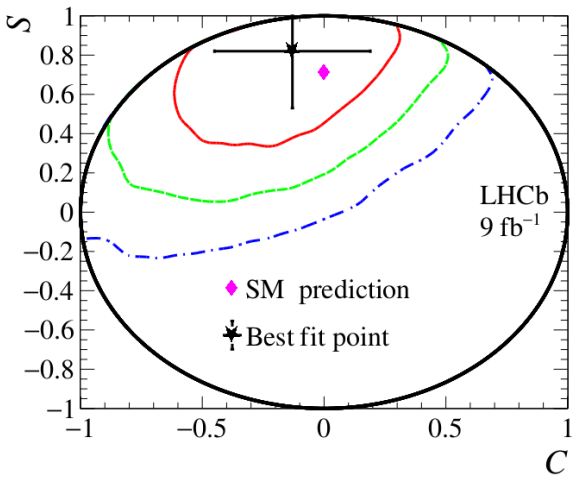

Figure 4: Two-dimensional confidence regions for C and S based on a likelihood-ratio ordering technique, corresponding to the total q^2 range. The red solid, green dashed and blue dash-dotted contours represent the 68.3%, 95.5%, and 99.7% confidence intervals given the physical constraint $C^2 + S^2 \leq 1$. The black marker with error bands represents the fit results for C and S. The magenta diamond marker stands for the SM prediction.

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